

Hints, Placebos and Shortcuts: Controlled Tests of Structural Primers for LLM Graph Reasoning

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Abstract

Prepending a *primer* of factual sentences about a graph’s structure (node degree, clustering coefficient, random-walk return probabilities or component count) to its text encoding is a natural way to assist a language model on graph-reasoning questions. We evaluate seven primer conditions against four controls jointly — a graph-blind shortcut solver refit at the evaluated graph size, a contentless `filler` preamble matched in length to the `clustering` primer, a minimal-length content control, and a degree-preserving rewiring — across four Qwen3 arms (1.7B and 4B, with and without thinking mode), six GraphQA tasks and seven edge densities on 40-node Erdős–Rényi graphs. A primer’s effect reverses across task, density and model size: the same `degree` primer is worth +26.5 points against the length control on `edge_count` and −6.0 on `node_degree`, where it states the requested number outright. What the primer changes is which procedure the model runs. Given a list of degrees, the share of responses that sum them and halve rises from 7% to 100%; where the primer states the answer, the median response collapses from 273 to 34 tokens and accuracy falls from 99 to 87. Accuracy also depends on where in the 40-line primer the queried node sits — by 24.3 points for the largest gradient we fit — though only 4 of 24 such gradients are distinguishable from the no-primer control’s when the difference is tested directly, and on a second task the control itself gains 25.0 points down the same 40 lines with no primer present; and bundling three primers into one recovers a median 0.50 of what they are worth separately, losing gains and damage alike. Two controls that are not usually applied decide how all of this reads. The first is a contentless preamble: it costs up to 15.7 points, but that is not a price for the tokens, because fitting the cost of added text across the five informative primers, which span 37 to 4,691 characters, gives −0.02 points per thousand characters. Length is close to free

here and this particular degenerate preamble is not. The second is balanced accuracy on the boolean task, where raw accuracy rises with density while the model converges on answering “yes” to 98–99% of pairs.

1 Introduction

Fatemi et al. (2023) show that a large language model’s accuracy on graph-reasoning questions varies substantially with the choice of text encoding when the graph and question are held fixed. We ask a complementary question about the same benchmark: does prepending a short *primer* of derived, node-level facts to the graph encoding improve the model’s answers?

Four confounds complicate the interpretation of such an effect. First, a primer may help because the model reads the answer from it rather than reasoning over the graph (answer leakage). Second, a primer changes prompt length, and context length and position alone affect retrieval and reasoning accuracy (Liu et al., 2024; Kamradt, 2023; Levy et al., 2024). Third, answer extraction can err in ways that correlate with response style, which appears as a spurious condition effect (Section 3.4). Fourth, a model near ceiling on a task — or one whose task has a majority class large enough to score well by itself — leaves no room for any primer effect to appear, or credits one that is not there.

We ask what a primer does, and the headings of Section 5 are the questions in order. Which primer moves which task? What does the primer change in the model’s response — how much it writes, and which procedure it states? Where in the primer does it read? Do primers combine? And which of the answers hold across model arms and corpora? Section 5.1 comes first and is prerequisite rather than a result: at $n=40$ two of the six tasks have a majority class that scores well by itself, one is bounded by the generation budget, and a contentless preamble is already worth up to 15.7

points even though added text of the same size costs nothing when it says something, so what counts as a primer effect has to be fixed before any of it can be read.

Two contributions support the answers. The first is the four controls of Figure 1 applied jointly — a graph-blind shortcut solver refit at the evaluated graph size (Section 3.2), a length-matched filler, a minimal-length components, and a degree-preserving rewiring (Section 3.3). The second is that we read the generated text as well as the score, which is what lets us say what a primer did rather than only how large its effect was.

2 Related Work

We adopt the GraphQA task definitions of [Fatemi et al. \(2023\)](#) and manipulate graph size and edge density directly, so that difficulty is an experimental variable rather than an incidental one. [Wang et al. \(2023\)](#) report that LLMs are brittle to spurious correlations in graph problems and propose *Build-a-Graph* and *Algorithmic* prompting, the prompt-level interventions closest to a primer; [Perozzi et al. \(2024\)](#) instead learn a continuous graph encoding. [Dziri et al. \(2023\)](#) argue that transformers solve compositional tasks by linearised subgraph matching rather than systematic reasoning, which raises the same question as a primer’s benefit: does it reflect reasoning or a shorter pattern to match? [Geirhos et al. \(2020\)](#) frame this as *shortcut learning*; our shortcut solver operationalises that concern for the primer channel. Irrelevant context degrades reasoning ([Shi et al., 2023](#)), as does input length well below the context limit ([Levy et al., 2024](#)); our *filler* condition measures this cost directly. The *rwse* condition adapts the random-walk structural encoding of [Dwivedi et al. \(2022\)](#), rendered as sentences, and the rewiring follows the double-edge-swap configuration model ([Fosdick et al., 2018](#)). We distinguish Qwen3’s native thinking mode, toggled through the chat template, from chain-of-thought *prompting* ([Wei et al., 2022](#)); this paper tests only the former.

3 Method

3.1 Primers

A primer is a preamble of factual sentences about a graph’s structure, prepended to the standard text encoding and the task question (Figure 1; a complete example appears in Appendix A.1). A single renderer produces all seven conditions, so they share

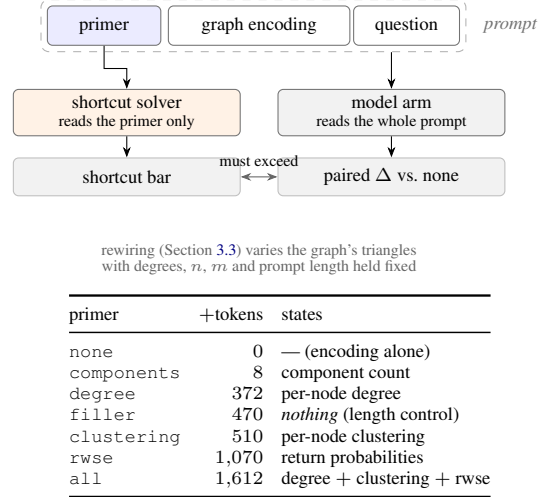


Figure 1: Experimental design. One prompt, two readers: the graph-blind shortcut solver sees only the primer and sets the bar a primer’s gain must exceed; the model arm sees all of it. Lengths are means at $n=40$ (Qwen3 tokenizer) against a mean *none* prompt of 2,165 tokens, so *filler* is a close length control for *clustering* but bounds only the length term for *rwse* and *all*.

sentence framing and number formatting and differ only in the facts stated. The exception is *filler*, which states no fact and therefore uses its own frame (“Node N is simply present within the graph G .”); it is a length control, not a content control. *rwse* states each node’s random-walk return probabilities after two and three steps ([Dwivedi et al., 2022](#)).

3.2 Shortcut solvers

A primer’s measured benefit is evidence of graph reasoning only if it exceeds what a solver that never sees the graph can recover from the primer text. Our shortcut solver combines sixteen exact rules, one heuristic, eight fitted rules (fit and scored on disjoint graph sets) and an exact enumeration bound for small graphs. Its accuracy on a (task, condition) cell, the *shortcut bar*, is a content-only lower bound: it never overstates what the primer text alone permits.

Bars are fit at the evaluated graph size: $n=40$ Erdős–Rényi graphs at the seven densities of the sweep, with disjoint 300-graph fit and test sets per density (Table 11); the screened arms, which answer the published 5–19-node split, use bars fit on that split, which differ substantially (Appendix Table 15). A (task, condition) pair offers a *substitute route* when its bar exceeds the *none* bar by more than 0.05; every gain is either at most 0.011 or at

least 0.066, so any threshold between these values yields the same split. Only `node_degree` (under `rwse`, `degree`, `all`) and `edge_count` (under `clustering`, `degree`, `all`) offer a route. `node_count` and `cycle_check` have constant gold answers at $n=40$ (40, and “yes” at every density) and are excluded.

3.3 Rewiring: isolating structure from length

`filler` controls length only approximately, and comparisons across graph generators confound structure with class balance, so we use a within-instance design. A double-edge swap replaces edges $(a, b), (c, d)$ with $(a, d), (c, b)$ (Fosdick et al., 2018), which preserves the degree sequence, n, m , the `node_degree` gold answer and the prompt length while changing the triangle count that the `clustering` primer reports. We generate low, base and high variants of each graph ($n=40$, mean degree 12, 200 graphs per level; mean clustering 0.01, 0.31, 0.73) and evaluate `node_degree` under `none`, `clustering` and `filler` with `qwen3-1.7b`, `qwen3-1.7b-think` and `qwen35-2b`. If the `clustering` benefit derives from the stated values, it should vary with triangle count. If instead the model reads mean clustering as a density prior (in an Erdős–Rényi graph $\bar{C} \approx p$), its degree estimates should shift toward $\bar{C}(n-1)$: 0.5, 12.0 and 28.7 across levels, against a true mean of 12.

3.4 Answer extraction and measurement validity

Extraction errors can masquerade as condition effects, so answers are extracted with task-specific patterns rather than a positional rule: on `cycle_check` a last-yes/no rule is not sufficient, because `qwen3-4b` often answers correctly and then recites that a cycle has “no repeated” nodes, which a positional rule scores against one condition more than another. The extractor runs on the raw response: nothing strips `<think>`, and 10–15% of thinking-arm generations never close the block, so their answer is read from inside it.

Truncated pairs are dropped, and that convention is consequential. Scoring truncated generations incorrect instead flips Benjamini–Hochberg significance in 21 of 144 (arm, task, condition) cells and moves nine by more than ten points; two `qwen3-4b-think/edge_count` effects reverse sign (−23.5 to +8.5, −41.5 to +1.7) and `qwen3-1.7b-think/degree` falls from

−61.5 to −3.2. All sit where the thinking arms truncate on 62–81% of generations, so few pairs survive. Table 1 reports n per cell and never bolds a cell below 100 pairs, because those survivors are selected on whether the generation terminated; the appendix gives both conventions for every cell. We read no thinking-arm `edge_count` result as evidence about primers.

3.5 Significance testing

Every comparison is paired on the same graph and question (Dror et al., 2018) and tested with the exact McNemar test (McNemar, 1947), Benjamini–Hochberg corrected (Benjamini and Hochberg, 1995) within each (arm, task) family at $q=0.05$; a claim that a condition beats both controls also requires significance under a global correction. For the effect–baseline relation (Section 5.6), baseline and effect are estimated from disjoint random halves of each cell’s graphs (Chernozhukov et al., 2018), and the route $\times$ baseline interaction is bootstrapped over (arm, task, density) blocks, which share one `none` sample. For null cells we simulate a minimum detectable effect (MDE) at power 0.8, $\alpha=0.05$.

4 Experimental Setup

Tasks and arms. The six GraphQA tasks (Fatemi et al., 2023) are those named in Table 1. An *arm* is a model checkpoint with a fixed decoding configuration, so native thinking mode off and on gives two arms of one checkpoint.

Why $n=40$. On the published GraphQA split (Fatemi et al., 2023) — 5–19 nodes, five tasks — eight arms from two families score 0.97–1.00 under `none` on four of the five tasks (Appendix Table 13), leaving no headroom for a primer to move. We therefore evaluate on graphs large and dense enough to leave room in both directions.

Main sweep. `Qwen3-1.7B` and `Qwen3-4B` (Qwen Team, 2025) carry it, each with thinking off (`qwen3-1.7b`, `qwen3-4b`) and on (`-think`): two sizes, both modes of the same weights, and the two sizes that retain headroom on five of six tasks at $n=40$. All four arms answer one shared prompt set of Erdős–Rényi graphs (Erdős and Rényi, 1959) with $n=40$ at densities $p \in \{.10, .20, .35, .50\}$, 100 graphs per (density, task) cell, seven conditions and six tasks, for 16,800 prompts per arm, extended on `node_degree` and `edge_existence` to

$p \in \{.65, .75, .85\}$ (100 graphs per cell, all four arms) so that the density range reaches from near-ceiling to near-floor. Decoding is greedy with a budget of 8,192 new tokens for every arm at $p \leq .50$. The plain arms’ high-density extension ran at 2,048 instead; responses there have a median of 256 new tokens and a 95th percentile of 410, and the lower budget truncated 8 of those 8,400 generations, so the two budgets are not interchangeable in principle but are in practice for that slice. Section 5.6 reports a comparison in which every arm is rescored at a common 2,048 tokens. Every headline result below is measured on this sweep.

Supporting corpora. Four further corpora each answer one question the main sweep cannot, and each is used only for that question. (i) A dedicated `node_degree` sweep with `qwen3-1.7b` and `qwen3-1.7b-think` — 400 graphs per density at all seven densities, with a `filler` arm, plus an independent replication on 400 newly seeded graphs per density at $p \leq .50$ — supplies the power the main sweep lacks for the one non-answer content effect (Section 5.6). (ii) A degree-preserving rewiring corpus (Section 3.3; `qwen3-1.7b`, `qwen3-1.7b-think` and `Qwen3.5-2B` (Qwen Team, 2026), `qwen35-2b`) tests whether that effect tracks the triangle count the primer reports. (iii) A fixed-mean-degree grid — mean degree ≈ 8 or ≈ 16 with $n \in \{20, 40, 80, 160\}$ on `node_degree`, answered by `qwen3-1.7b` and `qwen3-8b` — separates answer size from graph size and tests the same effect across four graph sizes (Section 5.6); it carries no `filler` arm. (iv) A 100-graph probe of `none` and `degree` on all six tasks, answered by `Qwen3-0.6B` with and without thinking and by `qwen35-2b`, adds a third family and a smaller size to the robustness checks of Section 5.6.

Screened arms. The eleven arms of the starting point and the probe support the two checks of Section 5.6; their shortcut bars are refit on their own corpus (Appendix Table 15).

Metrics. Exact match for every task; mean absolute error for the integer tasks is in the Appendix. On `connected_nodes` the choice of metric decides whether an effect can appear at all: `set-F1`, the metric that task is usually reported on, awards partial credit on a large answer set and sits at 0.93–1.00 in every cell, while exact match on the same responses spans 0.48–0.92. We report exact match

throughout. On `edge_existence` we report balanced accuracy alongside accuracy, because the share of node pairs that are joined rises from 10% to 85% across our density range (Section 5.1).

5 Results

5.1 What these tasks and scores measure

A primer can only move accuracy where accuracy can move, and on two of the six tasks the score moves for reasons that have nothing to do with the primer. This section fixes what the numbers in the rest of the paper mean.

Accuracy on `edge_existence` tracks the class prior. The share of node pairs that are joined rises from 10% at $p=.10$ to 85% at $p=.85$, so a constant answer earns a score that changes with density. Under `none`, `qwen3-1.7b` scores 89, 73, 66, 54, 68, 77, 86 across the seven levels — a decline and an apparent recovery. Balanced accuracy over the same responses runs 93.9, 83.5, 75.4, 55.8, 52.9, 52.1, 53.3: the recovery is entirely the prior, and above $p=.50$ the model answers “yes” to 98–99% of pairs. The same split holds on `qwen3-4b`, less severely (raw 96 to 96, balanced 97.8 to 89.4). We read no gain above $p=.50$ as evidence about graph reading (Appendix Figure 3). These are shifts in response bias: the hit rate stays at 98–100% under every condition, so accuracy moves only through false alarms, which `filler` raises and `clustering` lowers, and with hits at ceiling sensitivity and criterion are not separately identifiable (Macmillan and Creelman, 2005; Hautus, 1995) (Appendix Tables 17 and 18).

Two tasks have no label distribution at all. At $n=40$ every one of the 400 graphs contains a cycle and has 40 nodes, so `cycle_check` and `node_count` have a majority-class baseline of 1.000. `cycle_check` is uninformative here. `node_count` is worse: `qwen3-1.7b` scores 0.315 because 68.5% of its answers read the encoder’s “nodes 0, . . . , 39” preamble as the count, and every answer it gives is 39 or 40. Interposed text disturbs that off-by-one, which is what the large primer gains on this task measure — though not by length or by mere interposition, since the six conditions buy +2.0 to +67.3 and the ordering is not the length ordering. We exclude both tasks from every claim about graph reading.

Truncation is an outcome, not a missing value. `qwen3-1.7b` reaches the output budget on 22.4%

of `edge_count` generations; the thinking arms reach it on 81.0% and 62.2%. It decides scores elsewhere too: on `cycle_check`, whose answer is “yes” for all 400 graphs, `qwen3-1.7b` scores 0.997 on the generations that terminate and 0.923 counting the rest as wrong. Paired differences below drop a pair when either side truncated, which is not uniformly conservative — it favours the primer in some cells and `none` in others (Section 3.4); rates are in Table 16.

Difficulty is set by the answer, not the graph. `node_degree`, `connected_nodes` and `edge_existence` ask about the *same node of the same graph* and differ only in whether the answer is a count, a set or a boolean. Under `none` at $p=.50$, `qwen3-1.7b` scores 0.30, 0.48 and 0.54, and `qwen3-4b` reverses the ordering (0.99 against 0.79): the required output operation is a difficulty driver in its own right, though which operation is harder depends on the model and the 0.54 is a boolean at chance.

Added text is an active treatment, but length is not. `filler` states nothing about the graph and is not a null: it significantly reduces accuracy on `connected_nodes` and `edge_existence` in both plain arms. The cost is not a charge for tokens. It differs by task by a factor of two (Appendix Table 14a), is not monotone in density (running -5 , -20 , -32 , -6 , -2 , -1 , -1 from $p=.10$ to $.85$ on `qwen3-1.7b/edge_existence`) and changes sign within every task but `node_count`. Length itself is close to free: fitting Δ -versus-`none` on primer length across the five informative primers, which span 37 to 4,691 characters, gives -0.02 points per thousand characters, and `filler` sits 4.2 points below that line on average and 18.3 below at worst. So `filler` measures a degenerate preamble, not a token tax, and differences below are reported against it *and* against `none` because neither alone is the length control.

5.2 Which primer moves which task?

Tables 1 and 1 give the main sweep against both baselines. Three things are visible in them that a single headline would hide.

The sign is not a property of the primer. `degree` is worth $+26.5$ points against the length control on `qwen3-4b/edge_count` and -6.0 on the same arm’s `node_degree`, where the primer states the requested num-

ber outright. `clustering` is worth $+11.0$ on `qwen3-4b/edge_existence` and nothing on that arm’s `node_degree`. Between arms, `degree` on `node_degree` runs $+8.0$ on `qwen3-1.7b` and -6.0 on `qwen3-4b`. Effects reverse across task, across density and across model size, so no primer is good or bad in itself (Appendix Figure 5).

Effects are close to one-directional. The paired design decomposes each difference into items the primer fixed and items it broke, and the two are rarely both large. At $p=.50$, `degree` on `qwen3-4b/edge_count` fixes 33 items and breaks none; `rwse` on that arm’s `edge_existence` breaks 23 and fixes none; `degree` on its `node_degree` breaks 13 and fixes 1. `clustering` is the exception, fixing 8 and breaking 4 on `edge_existence`, so its small net is a different kind of effect from the others rather than a weaker version of the same one, and should not be read as one.

The largest content effect we measure costs eight tokens. `components` on `connected_nodes` gains $+9.2$ points over `none` and $+21.8$ over `filler` for `qwen3-4b` ($n=400$, $p \approx 10^{-4}$, both under the global correction; Table 1). Every condition beats `filler` there, because `filler` itself costs 12.5 points; what distinguishes `components` is that it also beats `none`. Since it adds 8 tokens where `filler` adds 470 (Figure 1), it is the only effect we measure that is substantially free of length cost.

Carrying the right feature is neither necessary nor sufficient. Each primer reports one graph feature and each task needs one, so some (primer, task) pairs are matched and the rest are not. The three largest gains we measure are matched pairs — `degree` on `edge_count` ($+26.0$ on `qwen3-4b`), `components` on `connected_nodes` ($+21.7$) and `clustering` on `edge_existence` ($+19.5$ on `qwen3-1.7b`). But mismatched content is still worth $+8.1$ points on `qwen3-1.7b`, more than two of the four matched pairs, and the fourth matched pair reverses with model size. Relevance organises the largest effects without predicting the rest (Appendix Table 10).

Arm	Task	none	<i>n</i>	Δ vs none						Δ vs filler			
				filler	comp.	clust.	rwse	degree	all	comp.	clust.	rwse	degree
qwen3-1.7b	edge_existence	70.5	399–400	−15.7	−8.2	+3.8	+5.3	+4.0	+11.5	+7.5	+19.5	+21.1	+19.8
	node_degree	60.5	399–400	−0.5	+1.8	+3.0	+4.0	+7.5	+10.3	+2.3	+3.5	+4.8	+8.0
	connected_nodes	72.5	399–400	−6.2	+0.8	+1.3	+1.0	+3.5	+2.0	+7.0	+7.5	+7.3	+9.8
	edge_count	1.8	211–253	−0.4	−1.4	−0.4	−0.8	+4.0	+2.4	−0.4	−0.4	−0.4	+4.5
qwen3-4b	edge_existence	90.5	398–400	−7.0	−2.8	+4.0	−10.0	−4.3	−9.8	+4.2	+11.0	−3.0	+2.5
	node_degree	99.2	400	−0.5	+0.2	−0.5	−2.3	−6.5	−8.0	+0.7	+0.0	−1.8	−6.0
	connected_nodes	86.0	400	−12.5	+9.3	+1.0	+0.0	−3.0	−2.7	+21.8	+13.5	+12.5	+9.5
	edge_count	2.0	393–400	+1.2	−1.3	−1.3	−1.5	+27.7	+3.8	−2.5	−2.5	−2.8	+26.5
qwen3-1.7b-think	edge_existence	99.2	397–399	−0.8	−0.8	−1.0	−1.3	−2.3	−2.0	+0.0	−0.3	−0.5	−1.5
	node_degree	76.8	376–396	−3.5	−1.0	+1.0	+2.5	+11.7	+11.6	+2.3	+4.5	+5.8	+14.6
	connected_nodes	79.7	390–394	+2.0	+2.8	+1.5	−0.3	+7.4	+6.1	+0.5	−0.5	−2.5	+4.6
	edge_count	86.3	<i>13–29</i>	+3.4	+7.4	−8.7	−4.0	−61.5	−8.3	+5.4	−4.2	+13.0	−18.8
qwen3-4b-think	edge_existence	100.0	393–399	+0.0	+0.0	+0.0	+0.0	+0.0	+0.0	+0.0	+0.0	+0.0	+0.0
	node_degree	99.7	380–388	+0.0	−0.3	−0.3	−2.1	+0.0	−0.8	−0.5	−0.3	−2.1	−0.3
	connected_nodes	98.7	373–379	+0.8	+0.5	+0.0	+0.0	+0.0	−0.5	+0.0	+0.0	−0.5	−0.5
	edge_count	97.6	<i>34–60</i>	−2.0	+1.7	+1.7	+0.0	−23.5	−41.5	+2.3	+0.0	−6.3	−19.3

Table 1: Main sweep, $n=40$, pooled over densities $p \in \{0.10, 0.20, 0.35, 0.50\}$, 100 graphs per (task, primer, density) cell before truncation. The `none` column is exact-match accuracy (%); n is the range, across the row, of pairs in which neither side truncated; the rest are paired differences in percentage points, against `none` on the left and against `filler` on the right, with truncated generations excluded pairwise. `filler` is length-matched to `clustering` only (Figure 1), so the right half is a length-matched contrast for that column alone. `node_count` and `cycle_check` are omitted: gold is constant at $n=40$, so a shift there is not graph reading (Section 5.1); both appear in Appendix Table 2 onward. **Bold**: significant, exact McNemar, Benjamini–Hochberg within each (arm, task) family and baseline, $q = 0.05$. A row whose n is *italic* falls below 100 surviving pairs and is never bolded, because those survivors are selected on whether the generation terminated (Section 3.4).

5.3 What does the primer change in the response?

Accuracy alone cannot say what a primer did. The generated text can, and it shows the primer changing which procedure the model runs and how much work it does, in directions that differ by task.

Before the primer, `edge_count` is budget-bound. Under `none` the plain arms almost never answer exactly — `qwen3-4b` is exact on 8 of 400 with a median relative error of 10.9% and never truncates, `qwen3-1.7b` exact on 5 with 28.5% over the 70% of its generations that terminate — while the thinking arms are exact on 86% and 98% of the generations that terminate and truncate on 82% and 80% of all of them. The task is not beyond the plain arms’ competence; without an explicit serial count they substitute an estimate, and with one they run out of budget.

On `edge_count` the primer replaces the method. Under `none` at $p=.50$, 7% of `qwen3-1.7b`’s responses state the handshake shortcut — sum the degrees and divide by two — and the rest enumerate edges. Under `degree`, which lists all 40 degrees, 100% state it. Truncation falls from 29% to 20% and the

median terminating response from 3,737 to 818 new tokens. The same switch appears at every density, and at $p=.10$ runs 67% to 99%. This is a wholesale change of algorithm, not a nudge: the primer supplies an arithmetic route around an enumeration the model cannot finish inside its budget. `filler` induces the switch in 31% of responses and `all` in 40%, so stating the degrees is what carries it.

Where the primer hurts, reasoning collapses.

On `qwen3-4b/node_degree` at $p=.50$, where the `degree` primer states the requested number verbatim, the median response falls from 273 new tokens under `none` to 34, and accuracy falls from 99 to 87 — the primer breaks 13 items and fixes 1. Neither cell truncates at all, so this is a choice and not a budget limit. `filler` leaves the median at 266 tokens and costs 2 points, so it is the stated answer and not the added text that shortens the response. The model abandons a reliable 273-token count for an unreliable 34-token lookup.

The direction is task-specific. On `qwen3-1.7b/edge_existence` the same primers make the model do *more*: 206 tokens under `none` against 294 under `degree` and 352 under

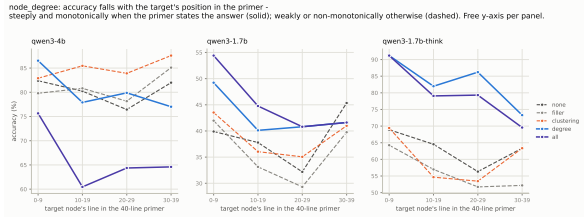


Figure 2: `node_degree` accuracy by the queried node’s line number within the 40-line primer, pooled over all seven densities ($n=647\text{--}700$ per condition). Primers that state the answer show the steepest gradients, but so does `rwse`, which never prints the queried degree, and the `filler` control is not flat either.

`all`. On `qwen3-4b/connected_nodes`, `filler` alone cuts the median from 233 to 94 while costing 12.5 points — a contentless preamble that makes the model answer faster and worse. “Primers make responses longer” and “primers make responses shorter” are both false as general statements (Appendix Table 8).

At high density the smaller model stops reasoning altogether. On `edge_existence` beyond $p=.50$, `qwen3-1.7b`’s responses shorten sharply and converge on the majority answer: it replies “yes” to 98–99% of pairs and its balanced accuracy sits at chance (Appendix Figure 3). Neither prompt length nor task difficulty triggers this. A ladder that raises n at fixed mean degree makes the prompt longer and the task harder without producing it, and the collapse appears on the task whose answer space is a single token while the enumeration tasks on the same graphs keep generating (Appendix Figure 4). What the two-valued answer format permits is a cheap guess that is right most of the time, and raw accuracy rewards it.

5.4 Where in the primer does the model read?

If a primer is being retrieved from rather than reasoned over, accuracy should depend on *where* in the 40 lines the queried node sits. It does, but most of that dependence is not the primer’s.

We fit correctness on the target’s line number and report the slope as points from the first line to the last, with a percentile bootstrap interval over the 647–700 items in each cell (Figure 2). The steepest are `all` on `qwen3-1.7b-think`, -24.3 $[-33.5, -15.2]$, and `degree` on the same arm, -18.7 $[-27.7, -9.7]$. But `none` slopes too, so the quantity to test is the *difference* (Gelman and Stern, 2006). A paired bootstrap of $\text{slope}(\text{condition}) - \text{slope}(\text{none})$ over the same graphs leaves

4 of 24 `node_degree` contrasts excluding zero, uncorrected: `rwse` (-23.0 $[-33.4, -12.0]$), `all` (-22.1 $[-34.5, -9.5]$) and `degree` (-13.8 $[-25.1, -2.5]$) on `qwen3-1.7b`, and `all` on `qwen3-1.7b-think` (-15.5 $[-28.1, -2.1]$). On `qwen3-4b`, `degree`’s excess over `none` is -10.8 $[-23.3, +1.4]$ — not distinguishable.

The control slopes, and on one task it slopes upward. `node_degree` and `connected_nodes` ask about the same node of the same graph, so the position variable is identical. Run the same fit on `connected_nodes` and `qwen3-4b` gains 25.0 points $[14.5, 36.7]$ down the primer under `none`, and 26.3 $[12.0, 40.0]$ under `filler` — a large positive gradient with no primer present, on the nodes that give a flat `none` on `node_degree`. Whatever the line number tracks is partly a property of the task and the node, not of the preamble.

The confound is the ordering, not the node.

Node difficulty does not explain it: over the 19,600 `node_degree` items a node’s id correlates $+0.019$ with its degree and $+0.020$ with its clustering, and Erdős–Rényi ids are assigned before edges are drawn. But the incident encoding also lists nodes in id order, so a node late in the primer is late in the encoding as well, and `none` carries that ordering too. No condition here breaks it; a primer whose line order is shuffled away from node-id order would, and we did not run one. Nor is the gradient only the cost of finding a stated value: the steepest `node_degree` excess is `rwse`, which never prints the queried degree. Two further limits: four quartiles cannot distinguish monotone decay from a lost-in-the-middle shape, and the slope is not monotone in density ($-9, -12, -24, -24, -12, -33, -18$ for `degree` on `qwen3-1.7b-think`), so we do not claim it scales with how much there is to search.

5.5 Do primers combine?

`all` renders `degree`, `clustering` and `rwse` in one primer, so the parts and the bundle are both measured on the same graphs, and `all` is not shorter than its parts (4,691 characters against 5,469 summed), so nothing is dropped to make room. Over the 41 cells where the parts do anything, 25 have `all` moving the same way as their sum but by less. The median ratio is 0.50 $[0.29, 0.71]$. The reference here is not exactly 1: cells enter on a noisy denominator, and selecting

on a noisy quantity inflates it, so a median of ratios sits below 1 even under exact additivity. That bias runs against the claim, so the direction — the bundle delivers less than its parts — survives it, and we read no coefficient off it (Appendix Table 9).

The bundle is closer to one part than to three. Two accounts fit a shortfall. Under dilution the useful sentence is buried among irrelevant ones and every part shrinks; under the alternative the model gets one primer’s worth out of the three and the rest is inert. These differ only where no single part dominates the sum, since otherwise “largest part” and “sum of parts” are the same number. On the 27 cells where the parts genuinely share the effect, the ratio to the *sum* is 0.52 [0.40, 0.71] while the ratio to the *largest single part* is 1.00 [0.74, 1.20]. What *all* delivers is indistinguishable from whichever of its three components does the most work on that cell, and less than their sum — but the same selection bias applies to both ratios, so this contrast is suggestive rather than a test.

The shrink is not measurably symmetric. A pure dilution account also predicts one shrink factor regardless of sign. Fitting *all* on the part-sum through the origin gives 0.47 [0.33, 0.59] where the parts help and 0.47 [0.30, 1.20] where they hurt. The intervals overlap, so these data establish neither that the factor differs nor that it is the same, and we do not read the symmetry either way.

Two limits bound all of this. In 10 of the 48 cells *all* moves *against* its parts rather than with them, so the shrunk-sum picture is a tendency across cells and not a rule within them; *edge_count* supplies most of these, and it is also the task whose gain rests on a single stated procedure that the bundle buries. And the design contains no pairwise conditions — no *degree+clustering*, no *degree+rwse* — so nothing interpolates between one primer and three. Distinguishing “the model uses the most useful sentence” from “the model uses whichever sentence it happens to read first” needs those conditions, or a bundle whose order is permuted, neither of which we ran.

5.6 What holds across arms and corpora?

Effects live in a window, and mostly they leak. Binning every cell by its own *none* accuracy locates where a primer can act at all (Table 19). Movement is confined to the middle of the range: between 0.25 and 0.75 a primer whose text determines the answer is worth +16 points and one car-

rying only side information +2 to +4, while above 0.90 — where 53% of the side-information cells sit — both are within a point of zero. Inside the window and excluding leakage, *all* gains +5.3 points, *clustering* +2.9, *rwse* +2.2, *degree* +2.0 and components −0.3. A primer buys a few points of genuine help in a narrow band of difficulty, and several times that when it states the answer.

We report the window and not a slope of effect on baseline, which is not identified here: cells near ceiling can only lose and cells near floor can only gain, and answer-carrying primers change more answers (16.5% of pairs against 9.2%), which alone produces a negative slope. Flipping a random subset of each cell’s *none* outcomes at its own discordance rate, so the primer carries nothing, yields a cross-fitted interaction of −10.9 to −12.9 — as large as the real data give.

One non-answer content effect replicates. *clustering* on *node_degree* is the only effect of a primer that neither states the answer nor merely pays for its length that we could replicate on a dedicated corpus. Pooled over $p=.35$ and $.50$ it exceeds *none* by +5.9 points (95% CI [2.1, 9.8]) and *filler* by a similar margin, with *filler* and *none* indistinguishable there. That density window was chosen after inspecting the first seed, and it is the top two of the seven levels (+0.3, +3.3, +7.5, +4.3, −2.0, +0.5, −0.5), so it is not a pre-specified contrast; across all seven levels the effect is +1.9 ($p=0.022$). What makes the result stand is that the window was then fixed and tested out of sample: an independent-seed replication on newly generated graphs gives +5.5. On the fixed-degree grid of Appendix Table 12 it gains +6.2 points ($n=3,191$; $p<10^{-4}$). Neither mechanism we can test explains it: under degree-preserving rewiring, mean clustering rises from 0.01 to 0.73 while the benefit stays flat — though no contrast on that corpus is itself significant, so it bounds little — and the signed error shifts by less than 0.3 where a density-prior account predicts up to 17 (Appendix Figure 6 and Table 22).

Thinking mode buys accuracy partly with budget. Thinking brings most cells to ceiling, and where it does not, truncation censors the task. Rescored at a common 2,048-token budget — exact under greedy decoding, and costing the plain arms at most 0.2 points — the 1.7B thinking arm still wins on *edge_existence* (97.8 against 70.5%)

but loses on `cycle_check` (38.0 against 90.7%; Appendix Table 23). Many main-sweep nulls are uninformative: `none` is already at floor or ceiling in 42% of `qwen3-1.7b`'s null cells and in all 33 of `qwen3-4b-think`'s. Where a gain could be detected, the median detectable size is 5.5 points for the 1.7B arms and 5.4 for 4B.

6 Conclusion

A primer's effect on GraphQA accuracy is not a property of the primer. It reverses across task, density and model size, and it is close to one-directional within a cell — fixing many items and breaking none, or the reverse. Where it helps, it mostly helps by stating the answer: binned by baseline accuracy, a primer whose text determines the answer is worth +16 points in the 0.25–0.75 band and one carrying only side information +2 to +4, while above 0.90 — where half the cells sit — neither does anything.

Two controls decide how that reads. A contentless preamble is itself worth up to 15.7 points, so an effect measured against `none` alone confounds content with whatever that preamble does — but not with length: fitting the cost of added text across the five informative primers, which span 37 to 4,691 characters, gives −0.02 points per thousand characters, and `filler` sits 4.2 points below that line on average. The placebo is not a token tax but a stimulus of its own. And on the boolean task, raw accuracy rises with density while balanced accuracy falls to chance.

The generated text says what the primer did. A primer listing the degrees replaces edge enumeration with an arithmetic shortcut; one stating the requested number collapses the response from 273 tokens to 34 and costs accuracy, breaking 13 items to fix 1. Position inside the primer moves accuracy by up to 24.3 points, but only 4 of 24 gradients differ from the no-primer control's, and on a second task that control gains 25.0 points down the same lines. Bundling three primers recovers a median 0.50 of their separate worth.

All claims are behavioural. Evaluations of prompt-level graph augmentations should report a contentless preamble *and* a length series across the informative conditions, a graph-blind solver applied to the reported effects and not only to define a split, a majority-class baseline for every task, and enough of the generated text to say which procedure changed.

Limitations

The main sweep covers two model sizes from one family; the screened arms extend the range but on a smaller corpus with fewer conditions and were not pre-registered. All graphs are Erdős–Rényi, in which local clustering is nearly uniform, so the `clustering` primer carries little node-specific information; the rewiring varies triangle count within this family but does not test other families. The `filler` control matches `clustering` to within 8% in tokens but is 59× longer than `components`, 26% longer than `degree` and much shorter than `rwse` and `all`, so the Δ -versus-`filler` column is a length-matched contrast for `clustering` only; the length fit of Section 5.1 is what controls the others. `filler` also names every node in id order, so it states the vertex set and is not information-free. Every prompt uses one encoding (`incident`), one template and one primer position, with greedy decoding. Several results rest on 400 or fewer instances per cell; an MDE is reported for every null result, but it does not replace a larger sample.

Three limits bear on the behavioural results specifically. The solution strategy a response states is identified by a single surface pattern over free text (divided by 2, sum of the degrees, `handshak`); it is not hand-labelled, and a model may use a procedure it does not narrate, so the strategy shares should be read as the rate at which a response *says* it took the arithmetic route rather than the rate at which it did. Position within the primer is confounded with node identity and with position in the encoding, which is also node-id ordered; `none` does not bound this, because it carries the same ordering, and on `connected_nodes` it slopes upward by 25.0 points (Section 5.4). The response-length collapse is attributed to the answer format by eliminating prompt length and task difficulty, not by manipulating the format directly.

Four further conditions would settle the main open questions: *shuffled-value* (a stricter placebo than `filler`), *corrupted* (to separate copying from computation), *single-fact* (to separate dilution from interference in the bundle) and a *reordered* primer, which is the one that matters most: it is the only way to break the position/identity/encoding-order confound, and without it no positional claim here is causal.

Ethics Statement

This work evaluates open-weight language models on synthetically generated Erdős–Rényi graphs. It involves no human subjects, no personal or identifying data and no annotation labour. We see no direct risk of harm from the findings; the principal cost is the GPU compute for approximately 200,000 generations.

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A Supplementary material

A.1 Example prompt

A clustering-condition node_degree prompt at $p=0.10$ (elisions marked ...). Other conditions replace the first paragraph; none omits it.

Node 0 has clustering coefficient 0.20. Node 1 has clustering coefficient 0.00. ... Node 39 has clustering coefficient 0.00.

G describes a graph among nodes 0, 1, 2, ..., 38, and 39.

In this graph:

Node 0 is connected to nodes 2, 6, 8, 12, 35.

Node 1 is connected to nodes 5, 20.

...

Node 39 is connected to nodes 15, 16, 18, 20.

Q: What is the degree of node 16?

A:

A.2 Per-task results

Arm	none	filler	comp.	clust.	rwse	degree	all
1.7b	31.5	+39.8	+21.2	+67.3	+2.0	+6.8	+47.0
4b	90.2	+9.3	+9.0	+8.8	+9.3	+8.0	+9.8
1.7b-t	98.6	+1.4	+0.9	+1.4	+1.4	+0.8	+1.4
4b-t	99.5	+0.5	+0.0	+0.5	+0.5	+0.5	+0.5

Table 2: node_count, main sweep, pooled over $p \in \{0.10, 0.20, 0.35, 0.50\}$. The none column is exact-match accuracy (%); the rest are paired differences against it in percentage points, truncated generations excluded pairwise. Gold is constant at $n=40$ on this task, so a shift need not reflect graph reading. **Bold:** exact McNemar, Benjamini–Hochberg within the arm, $q = 0.05$.

Arm	none	filler	comp.	clust.	rwse	degree	all
1.7b	99.7	+0.3	-0.6	+0.3	-7.5	-12.6	-1.1
4b	99.7	-0.3	-1.3	+0.3	+0.3	-1.5	-0.5
1.7b-t	100.0	+0.0	+0.0	+0.0	+0.0	-0.7	+0.0
4b-t	100.0	+0.0	+0.0	+0.0	+0.0	-0.8	+0.0

Table 3: cycle_check, main sweep, pooled over $p \in \{0.10, 0.20, 0.35, 0.50\}$. The none column is exact-match accuracy (%); the rest are paired differences against it in percentage points, truncated generations excluded pairwise. Gold is constant at $n=40$ on this task, so a shift need not reflect graph reading. **Bold:** exact McNemar, Benjamini–Hochberg within the arm, $q = 0.05$.

Arm	none	filler	comp.	clust.	rwse	degree	all
1.7b	70.5	-15.7	-8.2	+3.8	+5.3	+4.0	+11.5
4b	90.5	-7.0	-2.8	+4.0	-10.0	-4.3	-9.8
1.7b-t	99.2	-0.8	-0.8	-1.0	-1.3	-2.3	-2.0
4b-t	100.0	+0.0	+0.0	+0.0	+0.0	+0.0	+0.0

Table 4: edge_existence, main sweep, pooled over $p \in \{0.10, 0.20, 0.35, 0.50\}$. The none column is exact-match accuracy (%); the rest are paired differences against it in percentage points, truncated generations excluded pairwise. **Bold:** exact McNemar, Benjamini–Hochberg within the arm, $q = 0.05$.

Arm	none	filler	comp.	clust.	rwse	degree	all
1.7b	60.5	-0.5	+1.8	+3.0	+4.0	+7.5	+10.3
4b	99.2	-0.5	+0.2	-0.5	-2.3	-6.5	-8.0
1.7b-t	76.8	-3.5	-1.0	+1.0	+2.5	+11.7	+11.6
4b-t	99.7	+0.0	-0.3	-0.3	-2.1	+0.0	-0.8

Table 5: node_degree, main sweep, pooled over $p \in \{0.10, 0.20, 0.35, 0.50\}$. The none column is exact-match accuracy (%); the rest are paired differences against it in percentage points, truncated generations excluded pairwise. **Bold:** exact McNemar, Benjamini–Hochberg within the arm, $q = 0.05$.

Arm	none	filler	comp.	clust.	rwse	degree	all
1.7b	72.5	-6.2	+0.8	+1.3	+1.0	+3.5	+2.0
4b	86.0	-12.5	+9.3	+1.0	+0.0	-3.0	-2.7
1.7b-t	79.7	+2.0	+2.8	+1.5	-0.3	+7.4	+6.1
4b-t	98.7	+0.8	+0.5	+0.0	+0.0	+0.0	-0.5

Table 6: connected_nodes, main sweep, pooled over $p \in \{0.10, 0.20, 0.35, 0.50\}$. The none column is exact-match accuracy (%); the rest are paired differences against it in percentage points, truncated generations excluded pairwise. **Bold:** exact McNemar, Benjamini–Hochberg within the arm, $q = 0.05$.

Arm	none	filler	comp.	clust.	rwse	degree	all
1.7b	1.8	-0.4	-1.4	-0.4	-0.8	+4.0	+2.4
4b	2.0	+1.2	-1.3	-1.3	-1.5	+27.7	+3.8
1.7b-t	86.3	+3.4	+7.4	-8.7	-4.0	-61.5	-8.3
4b-t	97.6	-2.0	+1.7	+1.7	+0.0	-23.5	-41.5

Table 7: edge_count, main sweep, pooled over $p \in \{0.10, 0.20, 0.35, 0.50\}$. The none column is exact-match accuracy (%); the rest are paired differences against it in percentage points, truncated generations excluded pairwise. **Bold:** exact McNemar, Benjamini–Hochberg within the arm, $q = 0.05$.

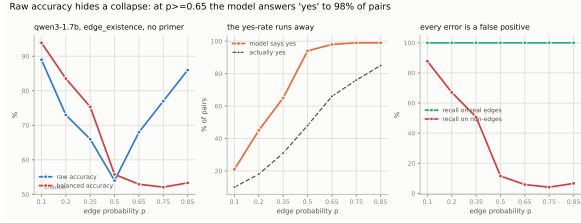


Figure 3: `edge_existence` under none, qwen3-1.7b. Accuracy falls to 54% at $p=.50$ and then rises again to 86% at $p=.85$, which reads as a recovery. Balanced accuracy over the same responses does not recover: it reaches chance at $p=.65$ and stays there, because the model answers “yes” to 98–99% of pairs while 66–85% of them really are edges.

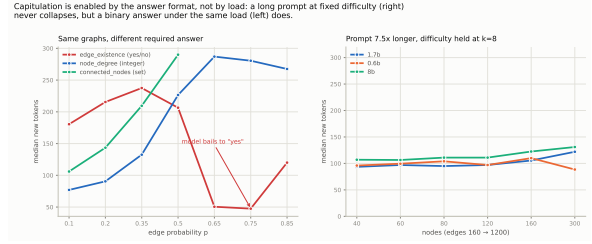


Figure 4: The response-length collapse on `edge_existence` and two controls on it. (a) median new tokens by density under none for three tasks answered about the *same* graphs: only the task with a two-valued answer collapses, while the tasks that must enumerate keep generating, so the trigger is not the graph. (b) median new tokens against n at fixed mean degree, where the prompt grows and the task gets harder without producing a collapse. What the collapse tracks is the answer format.

Arm / task		none	filler	degree	all
1.7b edge_count	median tokens	–	–	818	–
	% sum-of-deg.	7%	31%	100%	40%
	% truncated	29%	41%	20%	44%
4b node_degree	median tokens	273	266	34	134
	% sum-of-deg.	0%	0%	0%	0%
	% truncated	0%	0%	0%	0%
1.7b edge_existence	median tokens	206	171	294	352
	% sum-of-deg.	0%	0%	1%	0%
	% truncated	0%	0%	0%	0%
4b connected_nodes	median tokens	233	94	258	107
	% sum-of-deg.	0%	0%	0%	0%
	% truncated	0%	0%	0%	0%

Table 8: Response behaviour at $p=0.50$. Median new tokens is computed over generations that terminated, since a capped response is truncated at the budget rather than long, and is reported as – where more than 20% of generations truncated, because the surviving median is then a sample of the short responses; read the truncation row for those cells. “sum-of-deg.” is the share of responses that state the handshake shortcut (sum the degrees, divide by two) rather than enumerating edges. The primer moves length in both directions depending on the task, and on `edge_count` it replaces the method outright.

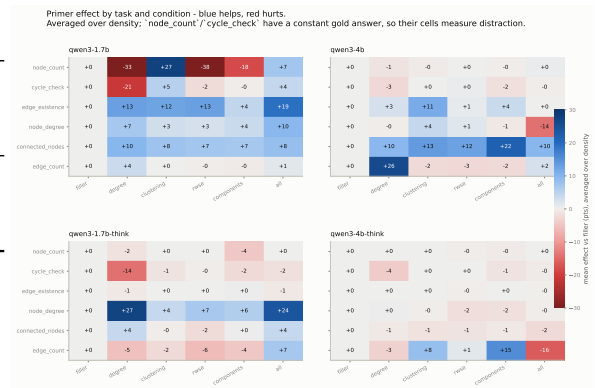


Figure 5: Every cell of Table 1 as a heat map: mean paired effect against filler by task and primer, one panel per arm, on a shared symmetric scale so that zero is the same colour throughout. The point of the figure is that the sign is not a property of a primer: rows and columns both change colour.

Arm	Task	$\sum$ parts	all	ratio
<i>other tasks</i>				
1.7b	connected_nodes	+5.7	+2.0	0.35
1.7b	edge_existence	+13.0	+11.5	0.88
1.7b	node_degree	+14.5	+10.2	0.71
1.7b-t	connected_nodes	+8.6	+6.1	0.71
1.7b-t	edge_existence	-4.5	-2.0	0.44
1.7b-t	node_degree	+15.6	+11.5	0.74
4b	edge_existence	-10.3	-9.8	0.95
4b	node_degree	-9.2	-8.0	0.86
<i>edge_count</i>				
1.7b-t	edge_count	-84.5	-22.2	0.26
4b	edge_count	+24.9	+3.8	0.15
4b-t	edge_count	-22.5	-41.1	1.82
<i>median over cells</i>			0.50	

Table 9: all renders degree, clustering and rwse in one primer, so the parts and the bundle are both measured. The first two columns are paired differences against none in percentage points, averaged over $p \in \{0.10, 0.20, 0.35, 0.50\}$; rows whose parts sum to under 4 points are omitted, since a ratio of two null effects is not informative. The final row is the median over the 41 individual (arm, task, density) cells rather than over the averaged rows above, because averaging numerator and denominator over densities first gives a ratio of means that the largest cells dominate. Its bootstrap interval is $[0.29, 0.71]$. We read this as a direction and not as a coefficient: the cells are screened on a noisy denominator, and selecting on a noisy quantity inflates it, so a median of ratios sits below 1 even under exact additivity. The bias runs against the claim, so the bundle falling short of its parts survives it; its size does not (Section 5.5). edge_count is the extreme case: there the bundle keeps almost nothing and in two arms moves against its parts.

A.3 Response behaviour

A.4 Where effects live

A.5 Robustness analyses

Arm	node_count	node_degree	edge_count	connected_nodes	cycle_check	all	cap
gemma4-12b	1.00	1.00	0.93	1.00	1.00	0.987	0/150
gemma4-12b-think	1.00	1.00	1.00	1.00	1.00	1.000	43/150
gemma4-e4b	1.00	1.00	0.96	1.00	0.97	0.986	5/150
gemma4-e4b-think	0.97	1.00	0.93	1.00	0.97	0.973	0/150
qwen3-8b	1.00	1.00	0.43	1.00	0.97	0.880	0/150
qwen3-8b-think	1.00	1.00	1.00	1.00	1.00	1.000	7/150
qwen3-14b	0.97	1.00	0.40	1.00	0.97	0.867	0/150
qwen3-14b-think	1.00	1.00	1.00	0.97	1.00	0.993	2/150

Table 13: The published GraphQA split (5–19 nodes, 30 graphs per cell) under none, exact match. Every arm scores 0.97–1.00 on four of the five tasks; only edge_count separates them, and only for the two plain Qwen arms (0.40–0.43 against 0.93–1.00 for the rest). cap counts generations that hit the token budget, excluded from the accuracies. This corpus cannot measure a primer effect, which is why the main sweep moves to $n=40$.

Primer / task	1.7b	4b	1.7b-t	4b-t
<i>matched: the primer carries what the task needs</i>				
clustering / edge_existence	+19.5	+11.0	-0.2	+0.0
components / connected_nodes	+7.0	+21.7	+0.5	-0.0
degree / edge_count	+4.9	+26.4	-18.8	-14.5
degree / node_degree	+8.0	-6.0	+15.0	-0.2
<i>mismatched, mean over all pairs</i>				
—	+8.2	+2.0	+2.3	-2.4

Table 10: Each primer reports one graph feature and each task needs one, so a pair is *matched* when the primer carries what the task asks for. Entries are paired differences against filler in percentage points, averaged over $p \in \{0.10, 0.20, 0.35, 0.50\}$; constant-gold tasks are excluded. The three largest gains we measure are all matched pairs, but matching is not sufficient and not necessary: mismatched content is still worth +8.2 points on qwen3-1.7b, more than two of the matched pairs, and the fourth matched pair reverses with model size. That pair is degree on node_degree, the one case where the primer states the requested number out-right: it gains 8 points on the 1.7B arm and loses 6 on the 4B arm (Section 5.4).

Task	none	comp.	clust.	rwse	degree	filler	all
node_count	1.00	1.00	1.00	1.00	1.00	1.00	1.00
edge_count	.030	.030	.104	.030	1.00	.030	1.00
node_degree	.144	.144	.144	.210	1.00	.144	1.00
connected_nodes	.000	.000	.000	.001	.001	.000	.008
edge_existence	.735	.735	.727	.554	.746	.735	.741
cycle_check	1.00	1.00	1.00	1.00	1.00	1.00	1.00

Table 11: Shortcut bars at $n=40$ (mean over seven densities): accuracy of a solver that reads only the rendered primer, never the graph. **Bold**: the primer content alone determines the answer. The unbolded 1.00 entries reflect a constant gold answer, not primer content.

n	$\bar{d}$	qwen3-1.7b			qwen3-8b	
		none	clust.	Δ	none	Δ
20	8	0.912	0.865	-4.8*	0.998	+0.2
40	8	0.728	0.770	+4.2	0.995	+0.0
80	8	0.645	0.750	+10.5*	0.965	-0.8
160	8	0.624	0.693	+6.9*	0.897	-2.8
20	16	0.575	0.740	+16.5*	0.985	-0.2
40	16	0.398	0.476	+7.8*	0.975	+1.0
80	16	0.225	0.323	+9.8*	0.895	-1.8
160	16	0.224	0.211	-1.3	0.802	-4.0
pooled		0.542	0.604	+6.2	$n=3191, p<10^{-4}$	

Table 12: Fixed-mean-degree sweep on `node_degree`: mean degree $\bar{d}$ is held while n varies, so density falls as the prompt lengthens. Growing n eightfold at $\bar{d}\approx 8$ costs 29 points and saturates after $n=80$; doubling $\bar{d}$ costs 33–42 points at every n . Δ is clustering against none, exact match. *: Benjamini–Hochberg at $q=0.05$ across the eight cells. This corpus has no filler arm, so Δ is a net effect. qwen3-8b is a ceiling control.

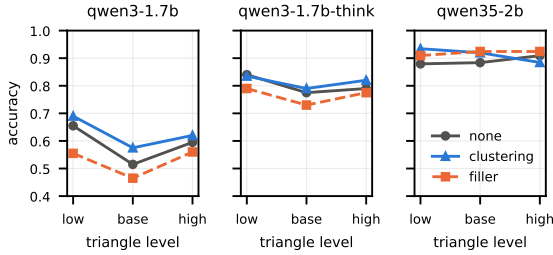


Figure 6: Rewiring: `node_degree` accuracy by triangle level (200 graphs per level; degree sequence and prompt length fixed; mean clustering 0.01, 0.31, 0.73).

A.6 Per-cell results

Paired effect sizes for every (arm, task, primer) cell: Δ in percentage points against none, an exact 95% interval conditional on the discordant pairs (Clopper–Pearson, inverting the same exact McNemar test used for the per-cell p -values), the permutation p -value, and Δ under the alternative convention that scores truncated generations incorrect rather than dropping the pair (Section 3.4). n is the number of pairs in which neither arm truncated. “n/a” marks a cell in which no pair disagreed: the conditional estimate is then exactly zero with no width, which is a statement about this sample and not a bound on the effect, so the discordant count is reported instead.

(a) length cost (filler–none) by task			
Task	cells	mean	range
edge_existence	11	-9.1	[-32.0, -1.0]
connected_nodes	7	-6.3	[-14.0, +3.1]
node_degree	11	-4.3	[-17.2, +4.0]
node_count	1	+31.0	[+31.0, +31.0]
(b) content gain (cond. – filler) by arm			
Cond.	1.7b / 1.7b-think / 4b		
components	+5.7 / +6.8 / +5.5		
clustering	+10.9 / +2.6 / +11.6		
rwse	+11.3 / +5.6 / +5.2		

Table 14: Mid-range cells. (a) The cost of added text is task-specific, not a per-token rate, and changes sign within tasks, so it cannot be transferred across them. `node_count` rests on a single cell and is a constant-gold task; it is shown for completeness only. (b) Only `components` holds its content gain across arms; the ranking of the other two reverses between 1.7b and 4b, so no single ordering of primers by length survives disaggregation.

Task	none	comp.	clust.	rwse	degree	filler	all
node_count	.064	.076	1.00	1.00	1.00	1.00	1.00
edge_count	.018	.018	.148	.018	1.00	.018	1.00
node_degree	.082	.082	.082	.616	1.00	.082	1.00
connected_nodes	.082	.082	.082	.082	.208	.082	.352
edge_existence	.498	.498	.720	.642	.794	.498	.794
cycle_check	.832	1.00	.832	.832	.946	.832	.946

Table 15: Shortcut bars on 500 graphs from the published 5–19-node generator, used for the screened arms. At $n=40$ (Table 11) the `edge_existence`/none bar rises to 0.735 through class imbalance, and degree’s gap over none on that task falls from +0.296 to +0.011.

Task	Primer	n	Δ	95% CI	perm p	zeroed
connected_nodes	components	400	+0.8	[-3.1, +4.5]	0.7861	+0.8
	clustering	400	+1.2	[-3.3, +5.7]	0.6458	+1.2
	rwse	400	+1.0	[-3.5, +5.5]	0.7304	+1.0
	degree	400	+3.5	[-0.8, +7.5]	0.1239	+3.5
	filler	400	-6.2	[-10.0, -1.9]	0.0040	-6.2
	all	399	+2.0	[-2.9, +6.8]	0.4644	+2.0
cycle_check	components	343	-0.6	[-0.6, +0.4]	0.5033	+2.0
	clustering	362	+0.3	[-0.3, +0.3]	1.0000	+8.5
	rwse	358	-7.5	[-8.1, -5.2]	0.0001	-0.2
	degree	301	-12.6	[-12.6, -10.3]	0.0001	-17.5
	filler	341	+0.3	[-0.3, +0.3]	1.0000	+3.0
	all	362	-1.1	[-1.6, +0.5]	0.2210	+7.0
edge_count	components	211	-1.4	[-2.3, +1.0]	0.3783	-1.0
	clustering	238	-0.4	[-2.4, +1.9]	1.0000	-0.5
	rwse	239	-0.8	[-2.3, +1.4]	0.6837	-0.8
	degree	253	+4.0	[+0.6, +5.8]	0.0179	+3.0
	filler	224	-0.4	[-2.0, +1.6]	1.0000	-0.5
	all	211	+2.4	[-1.1, +4.6]	0.2269	+0.8
edge_existence	components	400	-8.2	[-10.9, -4.7]	0.0001	-8.2
	clustering	400	+3.8	[-0.9, +8.1]	0.1199	+3.8
	rwse	399	+5.3	[+0.4, +9.8]	0.0321	+5.0
	degree	400	+4.0	[-0.7, +8.4]	0.0957	+4.0
	filler	400	-15.8	[-18.3, -12.0]	0.0001	-15.8
	all	400	+11.5	[+6.5, +15.8]	0.0001	+11.5
node_count	components	400	+21.2	[+15.9, +25.6]	0.0001	+21.2
	clustering	400	+67.2	[+65.4, +67.2]	0.0001	+67.2
	rwse	400	+2.0	[-4.5, +8.5]	0.5871	+2.0
	degree	400	+6.8	[+2.9, +9.9]	0.0008	+6.8
	filler	400	+39.8	[+37.2, +40.6]	0.0001	+39.8
	all	400	+47.0	[+43.8, +48.7]	0.0001	+47.0
node_degree	components	400	+1.8	[-2.4, +5.8]	0.4599	+1.8
	clustering	400	+3.0	[-2.0, +7.8]	0.2577	+3.0
	rwse	399	+4.0	[-0.9, +8.6]	0.1104	+4.0
	degree	400	+7.5	[+2.8, +11.6]	0.0018	+7.5
	filler	400	-0.5	[-4.6, +3.7]	0.9028	-0.5
	all	400	+10.2	[+4.9, +15.0]	0.0003	+10.2

Task	1.7B	1.7B-th.	4B	4B-th.
edge_count	22.4%	81.0%	0.6%	62.2%
cycle_check	6.1%	6.5%	1.0%	1.9%
node_degree	0.0%	1.1%	0.0%	1.6%
connected_nodes	0.0%	0.8%	0.0%	3.5%
edge_existence	0.0%	0.2%	0.1%	0.8%
node_count	0.0%	2.5%	0.0%	0.1%

Table 16: Truncation rate (share of main-sweep generations reaching the 8,192-token output budget), by task and arm, pooled over condition and density. **Bold:** above 50%.

Arm	Cond.	hit rate	false alarm
qwen3-1.7b	none	100.0%	40.3%
	filler	100.0%	61.8%
	clustering	100.0%	35.2%
qwen3-4b	none	99.1%	12.6%
	filler	100.0%	22.5%
	clustering	100.0%	7.5%

Table 17: edge_existence, pooled over the four main-sweep densities: hit rate (correct on gold “yes”) and false-alarm rate (“yes” on gold “no”). Accuracy changes almost entirely through the false-alarm rate.

Cond.	qwen3-1.7b				qwen3-4b			
	$\Delta d'$	Δc	$\Delta d'$	Δc	$\Delta d'$	Δc	$\Delta d'$	Δc
components	-0.36	[-0.56, -0.21]	-0.18	+0.07	[-0.24, +0.33]	-0.10		
clustering	+0.21	[+0.03, +0.41]	+0.11	+0.43	[+0.12, +0.75]	+0.08		
rwse	+0.07	[-0.19, +0.43]	+0.28	-0.38	[-0.68, -0.15]	-0.32		
degree	+0.15	[-0.05, +0.41]	+0.19	-0.04	[-0.33, +0.21]	-0.15		
filler	-0.93	[-1.16, -0.71]	-0.46	-0.25	[-0.54, +0.00]	-0.25		
all	+0.55	[+0.35, +0.77]	+0.27	-0.36	[-0.71, -0.11]	-0.31		

Table 18: edge_existence, main sweep: change in sensitivity d' and criterion c against none, averaged over the four densities (log-linear correction; 95% bootstrap CI for $\Delta d'$). With the hit rate at ceiling, d' and c are not separately identifiable.

Accuracy under none	primer carries		side information	
	cells	Δ	cells	Δ
0.00–0.25	19	+7.8	29	−1.8
0.25–0.50	9	+16.0	20	+2.3
0.50–0.75	9	+16.7	47	+3.9
0.75–0.90	4	−5.7	52	+0.6
0.90–1.00	28	−1.4	176	−0.8

side information only, baseline 0.25–0.90

components	mean -0.3, median -0.5 over 28 cells
clustering	mean +2.9, median +4.0 over 28 cells
rwse	mean +2.2, median +3.0 over 28 cells
degree	mean +2.0, median +1.5 over 18 cells
all	mean +5.3, median +3.0 over 17 cells

Table 19: Mean paired effect against none (percentage points) for every (arm, task, density, primer) cell of the main sweep, binned by the cell’s own none accuracy and split by whether the primer text alone determines the answer (shortcut bar ≥ 0.85 ; Table 11). Constant-gold tasks are excluded. Movement lives in the middle of the baseline range in both columns, and a primer that carries the answer is worth several times one that does not; above 0.90, where most cells sit, neither does anything.

Comparison	Δ	n	density	p
clust. vs. none (main)	+3.0	400	.10–.50	0.26
clust. vs. none	+1.9	2800	.10–.85	0.022
filler vs. none	-2.4	2800	.10–.85	0.0013
clust. vs. filler	+4.3	2800	.10–.85	2.4×10^{-8}
clust. vs. none	+7.5	400	.35	0.0089
clust. vs. none (new seed)	+4.2	1600	.10–.50	0.00055

Table 20: qwen3-1.7b, node_degree. All rows except the first come from the dedicated 400-graph-per-density sweep; the last row uses newly seeded graphs. Per-density values are in Appendix Table 21.

Task	Primer	n	Δ	95% CI	perm p	zeroed
connected_nodes	components	392	+2.8	[-0.7, +5.9]	0.1268	+2.5
	clustering	394	+1.5	[-2.7, +5.6]	0.5384	+2.0
	rwse	393	-0.3	[-4.3, +3.8]	1.0000	+0.2
	degree	390	+7.4	[+3.5, +10.6]	0.0001	+6.2
	filler	394	+2.0	[-1.7, +5.5]	0.3275	+2.5
	all	392	+6.1	[+2.2, +9.4]	0.0027	+6.2
cycle_check	components	346	+0.0	n/a (0 disc.)	1.0000	-7.5
	clustering	376	+0.0	n/a (0 disc.)	1.0000	-0.5
	rwse	386	+0.0	n/a (0 disc.)	1.0000	+2.2
	degree	295	-0.7	[-0.7, +0.5]	0.5036	-21.8
	filler	388	+0.0	n/a (0 disc.)	1.0000	+3.0
	all	373	+0.0	n/a (0 disc.)	1.0000	-0.8
edge_count	components	27	+7.4	[-5.1, +7.4]	0.5037	-0.8
	clustering	23	-8.7	[-17.2, +10.6]	0.6251	-1.2
	rwse	25	-4.0	[-4.0, +3.8]	1.0000	-3.2
	degree	13	-61.5	[-61.5, -16.1]	0.0102	-3.2
	filler	29	+3.4	[-8.4, +10.2]	1.0000	-1.5
	all	24	-8.3	[-22.8, +13.9]	0.6794	+5.0
edge_existence	components	398	-0.8	[-1.9, +0.9]	0.5090	-0.8
	clustering	398	-1.0	[-1.9, +0.6]	0.2911	-1.0
	rwse	398	-1.3	[-2.4, +0.6]	0.2274	-1.2
	degree	397	-2.3	[-3.1, -0.3]	0.0249	-2.5
	filler	399	-0.8	[-1.9, +0.9]	0.5051	-0.5
	all	399	-2.0	[-2.9, -0.1]	0.0374	-1.8
node_count	components	342	+0.9	[-0.6, +1.4]	0.3758	+1.5
	clustering	369	+1.4	[-0.1, +1.4]	0.0597	+9.0
	rwse	369	+1.4	[-0.1, +1.4]	0.0597	+9.0
	degree	360	+0.8	[-0.8, +1.8]	0.4512	+5.5
	filler	369	+1.4	[-0.1, +1.4]	0.0597	+9.0
	all	369	+1.4	[-0.1, +1.4]	0.0597	+9.0
node_degree	components	395	-1.0	[-4.4, +2.6]	0.6543	-1.0
	clustering	396	+1.0	[-3.0, +4.9]	0.7010	+1.5
	rwse	394	+2.5	[-1.9, +6.8]	0.2898	+2.5
	degree	376	+11.7	[+7.4, +15.0]	0.0001	+8.8
	filler	396	-3.5	[-7.3, +0.6]	0.0954	-3.0
	all	396	+11.6	[+7.3, +15.0]	0.0001	+12.2

Table 26: qwen3-1.7b-think: all cells.

p	clust.—none	filler—none	clust.—filler
.10	+0.3	+3.0*	−2.8*
.20	+3.3	+0.5	+2.8
.35	+7.5*	+1.0	+6.5*
.50	+4.3	−2.0	+6.3*
.65	−2.0	−9.5*	+7.5*
.75	+0.5	−6.3*	+6.8*
.85	−0.5	−3.5*	+3.0*

Table 21: qwen3-1.7b, node_degree, dedicated sweep, 400 graphs per density. *: McNemar $p < 0.05$ (uncorrected).

Arm	low	base	high
predicted shift	−11.5	0.0	+16.7
qwen3-1.7b	+0.03	+0.28	−0.02
qwen3-1.7b-think	+0.01	+0.04	+0.01
qwen35-2b	−0.29	−0.26	−0.16

Table 22: Density-prior test on the rewiring corpus: mean paired shift in signed node_degree error, clustering minus none (errors clipped at ± 20 , 200 graphs per level). “Predicted” is the shift if answers moved fully to $\bar{C}(n-1)$.

Task	1.7B, acc. under none (%)			4B, acc. under none (%)		
	plain	think	think@2048	plain	think	think@2048
connected_nodes	72.5	78.7	67.2	86.0	95.5	89.0
cycle_check	90.7	97.0	38.0	99.0	99.3	62.0
edge_count	1.3	15.8	0.0	2.0	20.0	0.0
edge_existence	70.5	99.0	97.8	90.5	99.8	99.8
node_count	31.5	91.0	86.0	90.2	99.5	94.8
node_degree	60.5	76.2	67.7	99.3	97.0	91.2

Table 23: Budget-matched comparison. Every arm ran at 8,192 tokens; “think” is that run and “think@2048” rescores it at 2,048, which is exact under greedy decoding. The plain columns are the 8,192 run; rescoreing them at 2,048 moves them by at most 0.2 points, because the plain arms exceed that budget on almost no terminating generation, so the comparison is matched in effect. Truncated generations are scored incorrect here, so values differ from the dropped-pair convention of Table 1.

Arm	Task	none	comp.	clust.	rwse	degree	filler	all
1.7B	node_count	0.69	0.47	0.01	0.67	0.62	0.29	0.22
	edge_count	133.4	98.1	105.7	79.0	47.7	181.2	94.2
	node_degree	0.94	0.91	0.78	0.91	0.67	1.27	0.91
	node_count	0.10	0.01	0.01	0.01	0.02	0.01	0.00
4B	edge_count	39.2	40.1	43.4	56.0	40.4	29.3	28.1
	node_degree	0.02	0.01	0.02	0.05	0.14	0.02	0.49

Table 24: Mean absolute error on the integer tasks, plain arms, main sweep (non-truncated rows).

Task	Primer	n	Δ	95% CI	perm p	zeroed
connected_nodes	components	400	+9.2	[+6.1, +11.3]	0.0001	+9.2
	clustering	400	+1.0	[−3.1, +5.0]	0.6893	+1.0
	rwse	400	+0.0	[−4.2, +4.2]	1.0000	+0.0
	degree	400	−3.0	[−7.1, +1.4]	0.1882	−3.0
	filler	400	−12.5	[−16.2, −7.9]	0.0001	−12.5
	all	400	−2.8	[−7.0, +1.7]	0.2511	−2.8
cycle_check	components	391	−1.3	[−1.8, +0.3]	0.1318	−2.0
	clustering	395	+0.3	[−0.2, +0.3]	1.0000	+0.5
	rwse	392	+0.3	[−0.2, +0.3]	1.0000	−0.2
	degree	391	−1.5	[−1.5, −0.1]	0.0312	−2.2
	filler	392	−0.3	[−0.3, +0.2]	1.0000	−0.5
	all	395	−0.5	[−1.0, +0.6]	0.6269	−0.2
edge_count	components	397	−1.3	[−2.1, +0.5]	0.1756	−1.2
	clustering	399	−1.3	[−2.4, +0.6]	0.2298	−1.2
	rwse	398	−1.5	[−2.0, +0.1]	0.0668	−1.5
	degree	393	+27.7	[+24.5, +29.4]	0.0001	+27.3
	filler	400	+1.2	[−0.9, +2.9]	0.3012	+1.2
	all	397	+3.8	[+1.1, +5.6]	0.0049	+3.8
edge_existence	components	400	−2.8	[−5.8, +0.7]	0.1254	−2.8
	clustering	400	+4.0	[+0.7, +6.7]	0.0168	+4.0
	rwse	400	−10.0	[−12.9, −6.2]	0.0001	−10.0
	degree	398	−4.3	[−7.3, −0.7]	0.0200	−4.8
	filler	400	−7.0	[−10.0, −3.3]	0.0002	−7.0
	all	399	−9.8	[−12.9, −5.8]	0.0001	−10.0
node_count	components	400	+9.0	[+6.9, +9.5]	0.0001	+9.0
	clustering	400	+8.8	[+6.2, +9.9]	0.0001	+8.8
	rwse	400	+9.2	[+6.9, +10.1]	0.0001	+9.2
	degree	400	+8.0	[+5.1, +9.7]	0.0001	+8.0
	filler	400	+9.2	[+7.5, +9.2]	0.0001	+9.2
	all	400	+9.8	[+8.0, +9.8]	0.0001	+9.8
node_degree	components	400	+0.2	[−0.2, +0.2]	1.0000	+0.2
	clustering	400	−0.5	[−1.7, +1.0]	0.7246	−0.5
	rwse	400	−2.2	[−3.1, −0.3]	0.0240	−2.2
	degree	400	−6.5	[−7.7, −4.0]	0.0001	−6.5
	filler	400	−0.5	[−1.7, +1.0]	0.7232	−0.5
	all	400	−8.0	[−8.9, −5.6]	0.0001	−8.0

Table 27: qwen3-4b: all cells.

Task	Primer	n	Δ	95% CI	perm p	zeroed
connected_nodes	components	377	+0.5	[−0.6, +1.0]	0.6249	+0.5
	clustering	373	+0.0	[−1.2, +1.2]	1.0000	−0.8
	rwse	379	+0.0	[−1.4, +1.4]	1.0000	+0.5
	degree	376	+0.0	[−1.5, +1.5]	1.0000	+0.5
	filler	379	+0.8	[−0.6, +1.3]	0.3787	+1.2
	all	375	−0.5	[−1.8, +1.1]	0.7306	−1.2
cycle_check	components	381	+0.0	n/a (0 disc.)	1.0000	−3.2
	clustering	394	+0.0	n/a (0 disc.)	1.0000	+0.0
	rwse	395	+0.0	n/a (0 disc.)	1.0000	+0.2
	degree	375	−0.8	[−0.8, +0.3]	0.2514	−5.5
	filler	396	+0.0	n/a (0 disc.)	1.0000	+0.5
	all	392	+0.0	n/a (0 disc.)	1.0000	−0.5
edge_count	components	60	+1.7	[−1.6, +1.7]	1.0000	+24.8
	clustering	59	+1.7	[−4.1, +5.0]	1.0000	+24.8
	rwse	49	+0.0	[−4.0, +4.0]	1.0000	+10.2
	degree	34	−23.5	[−23.5, −6.1]	0.0073	+8.5
	filler	50	−2.0	[−2.0, +1.9]	1.0000	+12.2
	all	41	−41.5	[−41.5, −25.3]	0.0002	+1.8
edge_existence	components	396	+0.0	n/a (0 disc.)	1.0000	−0.5
	clustering	396	+0.0	n/a (0 disc.)	1.0000	−0.8
	rwse	393	+0.0	n/a (0 disc.)	1.0000	−1.2
	degree	398	+0.0	n/a (0 disc.)	1.0000	+0.0
	filler	399	+0.0	n/a (0 disc.)	1.0000	+0.0
	all	394	+0.0	n/a (0 disc.)	1.0000	−1.0
node_count	components	400	+0.0	[−0.9, +0.9]	1.0000	+0.0
	clustering	399	+0.5	[−0.3, +0.5]	0.4936	+0.2
	rwse	399	+0.5	[−0.3, +0.5]	0.5005	+0.2
	degree	399	+0.5	[−0.3, +0.5]	0.5005	+0.2
	filler	400	+0.5	[−0.3, +0.5]	0.4870	+0.5
	all	400	+0.5	[−0.3, +0.5]	0.4870	+0.5
node_degree	components	380	−0.3	[−0.3, +0.3]	1.0000	+0.2
	clustering	386	−0.3	[−0.8, +0.6]	1.0000	+1.8
	rwse	381	−2.1	[−2.1, −0.5]	0.0072	−1.2
	degree	388	+0.0	[−0.5, +0.5]	1.0000	+2.5
	filler	386	+0.0	n/a (0 disc.)	1.0000	+2.0
	all	383	−0.8	[−0.8, +0.3]	0.2466	+0.5

Table 28: qwen3-4b-think: all cells.